Chapter 19

Missing Data in Mass Spectrometry: Mechanisms, Imputation Theory, and Sensitivity Auditing

Lucas VHH TRAN[1]

Abstract

Quantitative mass spectrometry (MS) datasets are intrinsically sparse. In label-free proteomics (LFQ), data-dependent acquisition (DDA), and untargeted metabolomics, between 20% and 50% of the data matrix cells are typically unobserved. In clinical biomarker cohorts, this missingness presents an acute methodological dilemma: standard multivariate algorithms (such as Principal Component Analysis, Hierarchical Clustering, and Support Vector Machines) require complete matrices, yet the numerical substitution of missing values alters the variance structure, covariance relationships, and downstream statistical inferences of the entire dataset.

Crucially, missingness in mass spectrometry does not arise from a single homogeneous process. A feature may be absent due to stochastic precursor selection during instrument duty cycles (Missing at Random; MAR), physical left-censoring below the instrument limit of detection (Missing Not at Random; MNAR), or true biological non-expression (e.g., tissue-specific proteins or gene knockouts). Applying a MAR algorithm (such as k-nearest neighbors) to an MNAR peptide invents artificial intermediate abundances for molecules that were genuinely absent, while applying left-censored substitution to MAR values introduces downward bias and artificially inflates differential abundance discoveries. This chapter establishes the rigorous statistical, biophysical, and computational foundations required to diagnose, model, and impute missing data in mass spectrometry. We formalize Rubin's missing data taxonomy within an analytical chemistry context, analyze the mathematical properties of deterministic, stochastic, and machine-learning imputation algorithms, derive Rubin's pooling rules for multiple imputation, and evaluate imputation-free probabilistic dropout modeling (proDA). The chapter concludes with a tutor diagnostic heuristics checklist, an end-to-end reproducible R laboratory, the "Five Deadly Sins" of omics imputation, and multi-tier exercises with exhaustive step-by-step worked solutions.

Keywords: Missing Data, Imputation, Missing Not At Random (MNAR), Missing At Random (MAR), Left-Censoring, QRILC, Multiple Imputation, Rubin's Rules, Sensitivity Analysis, proDA.

[1]CK-CARE Davos, Christine Kühne – Center for Allergy Research and Education, Davos, Switzerland. Email: lucas.tran@ck-care.ch

19.1 Tutor's Methodological Roadmap for Missing Data Management

Before applying ad-hoc imputation functions or running automated pipelines, the analytical chemist and biostatistician must diagnose the underlying missingness architecture. Table 19.1 outlines the four sequential checkpoints required to handle missing data rigorously.

Table 19.1: Tutor's Methodological Roadmap: Four Checkpoints for Missing Data Management

Stage	Physical Mechanism	Key Analytical Objective	Primary Failure Mode & Remedy
1. Diagnosis	Instrumental left-censoring ($y <$ LOD) vs. stochastic precursor selection [20, 16].	Fit logistic missingness curve (Eq. 19.5) to separate MNAR from MAR.	Misclassifying MNAR as MAR. Remedy: Test logistic slope γ_1; inspect intensity-dependent missingness [1].
2. Filtering	Extreme feature sparsity across biological replicates [29, 15].	Enforce 70% presence rule *within at least one biological group*.	Filtering across entire cohort discards true on/off biomarkers. Remedy: Filter group-wise.
3. Modeling	Tail-censored Gaussian draws vs. donor matching vs. probabilistic dropout [26, 1].	Substitute missing values appropriately, or model dropouts directly without imputation.	Deterministic zero/min replacement spikes variance. Remedy: QRILC or proDA [1].
4. Sensitivity Audit	Variance shrinkage underestimation from single imputation [21, 17].	Quantify inferential stability across multiple imputation draws via Rubin's rules.	Downstream p-values falsely anti-conservative. Remedy: Report multiple imputation pooled SE [28].

Methodological Note

Tutor's Note: The Golden Rule of Group-Wise Filtering
Never apply global missingness filters across an entire clinical cohort (e.g., "remove all proteins with $> 30\%$ missing values overall"). If a biomarker is abundant in Disease (100% detected in 10 patients) but silenced in Control (0% detected in 10 patients), global filtering eliminates this prime candidate. Always require $\geq 70\%$ detection *within at least one condition* [30, 29].

19.2 Biophysical Taxonomies of Missingness in Mass Spectrometry

In statistical theory, Donald Rubin [20] established the formal mathematical classification of missing data mechanisms based on the conditional distribution of the missingness indicator matrix.

19.2.1 Rubin's Missing Data Taxonomy

Let $\mathbf{Y} \in \mathbb{R}^{p \times n}$ denote the complete data matrix of p molecular features across n biological samples. We partition $\mathbf{Y} = (\mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}})$, where $\mathbf{Y}_{\text{obs}}$ represents observed quantitative intensities and $\mathbf{Y}_{\text{mis}}$ represents missing entries. Define the binary missingness response indicator matrix $\mathbf{M} \in \{0,1\}^{p \times n}$ such that:

$$M_{ij} = \begin{cases} 1 & \text{if } y_{ij} \text{ is missing} \\ 0 & \text{if } y_{ij} \text{ is observed} \end{cases} \tag{19.1}$$

The missing data mechanism is characterized by the conditional probability distribution $P(\mathbf{M} \mid \mathbf{Y}, \phi)$, parameterized by ϕ:

1. **Missing Completely at Random (MCAR):** Missingness is entirely independent of both observed and unobserved feature abundances:

$$P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}}, \phi) = P(\mathbf{M} \mid \phi) \tag{19.2}$$

   In MS instrumentation, MCAR occurs during random electrospray droplet instabilities, momentary liquid chromatography pressure spikes, or random ion guide transmission dropouts.

2. **Missing at Random (MAR):** Missingness depends systematically on observed data, but is conditionally independent of the unobserved missing value itself:

$$P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}}, \phi) = P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \phi) \tag{19.3}$$

   In DDA mass spectrometry, MAR arises from stochastic precursor selection: a peptide is present at moderate or high abundance, but competitive top-N precursor scheduling selects a co-eluting isobaric ion, leaving the peptide unfragmented.

3. **Missing Not at Random (MNAR):** Missingness depends directly on the unobserved value $\mathbf{Y}_{\text{mis}}$:

$$P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}}, \phi) \neq P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \phi) \tag{19.4}$$

   This is the primary physical driver of missingness in analytical chemistry: analytes present below the instrument limit of detection (LOD) or limit of quantitation (LOQ) fail to generate a discernible chromatographic peak above baseline noise.

19.2.2 Instrumental Censoring vs. Biological Non-Expression

A critical distinction must be drawn between analytical detection limits and biological absence:

- **Instrumental Left-Censoring (Analytical MNAR):** The protein or metabolite is physically expressed in the biological specimen, but its concentration falls below the ionization efficiency or detector sensitivity threshold ($y_{ij} < \text{LOD}$).
- **Biological Non-Expression (Structural Zero):** The gene is epigenetically silenced, unexpressed in the target tissue, or knocked out. The physical molecule count is identically zero ($y_{ij} \equiv 0$).

Methodological Pitfall & Deadly Sin

The Tissue-Atlas Imputation Catastrophe:
In multi-organ proteomics atlases (e.g., PXD010154 profiling 33,000 proteins across 12 human tissues), a liver-specific enzyme absent in cardiac muscle is missing due to biological regulation, not instrument failure. Imputing this enzyme in cardiac samples using left-censored algorithms introduces false molecular noise and destroys biological specificity. Never impute structural biological absence [16].

19.2.3 Mathematical Diagnosis of Missingness Mechanisms

To diagnose whether a dataset is dominated by MAR or MNAR, we fit a logistic regression model linking feature missingness probability to its mean observed abundance:

$$\text{logit}\left(P(M_{ij}=1 \mid \bar{y}_i^{\text{obs}})\right) = \ln\left(\frac{P(M_{ij}=1)}{1-P(M_{ij}=1)}\right) = \gamma_0 + \gamma_1 \bar{y}_i^{\text{obs}} \tag{19.5}$$

where $\bar{y}_i^{\text{obs}} = \frac{1}{|O_i|}\sum_{j\in O_i} y_{ij}$ denotes the mean observed $\log_2$ intensity for feature i.

- If $\gamma_1 \ll 0$ (strong negative slope; $p < 0.001$), missingness is heavily left-censored and abundance-dependent ($\implies$ **MNAR**).
- If $\gamma_1 \approx 0$ (flat slope across the dynamic range), missingness is abundance-independent ($\implies$ **MAR / MCAR**).

19.2.4 Imputation-Free Modeling: Probabilistic Dropout Analysis (proDA)

A modern paradigm shift championed by Ahlmann-Eltze and Huber [1] avoids artificial value substitution altogether. The `proDA` algorithm formulates the probability of missingness via a smooth sigmoidal dropout curve:

$$P(M_{ij}=1 \mid y_{ij}) = \frac{1}{1+\exp\left(\frac{y_{ij}-\rho_j}{\zeta_j}\right)} \tag{19.6}$$

where ρ_j is the 50% inflection point (approximate LOD) and ζ_j governs the slope of the detection limit in sample j. By integrating this likelihood directly into the empirical Bayes linear model, `proDA` calculates posterior abundance and moderated p-values without corrupting variance structures through single imputation [1].

19.3 Theoretical Formulations of Imputation Algorithms

19.3.1 Deterministic Minimum Value Imputation (MinDet)

Deterministic substitution replaces all missing entries with the minimum observed value within the sample:

$$y_{ij}^{*} = \min_{k\in O_j}(y_{kj}) \tag{19.7}$$

While simple, MinDet introduces an artificial mathematical singularity: if 20% of features receive identical values, the sample variance collapses, the empirical distribution exhibits a point-mass spike at $\min(y)$, and differential testing generates anti-conservative p-values.

19.3.2 Left-Censored Stochastic Draws (Perseus & QRILC)

To preserve the continuous variance structure of analytical left-censoring, Tyanova et al. [26] formulated the left-shifted truncated Gaussian model. For sample column j with observed mean $\hat{\mu}_j$ and standard deviation $\hat{\sigma}_j$, missing values are drawn stochastically:

$$y_{ij}^{*} \sim \mathcal{N}\left(\hat{\mu}_j - \delta\hat{\sigma}_j,\ (\omega\hat{\sigma}_j)^2\right) \tag{19.8}$$

where canonical parameters are $\delta = 1.8$ standard deviations (downward shift) and $\omega = 0.3$ (distribution width shrinkage).

Table 19.2: Mathematical classification and domain assumptions of core MS imputation methodologies.

Method	Algorithm Class	Core Mathematical Mechanism	Primary Failure Mode
MinDet / Zero	Deterministic Single	$y^*_{ij} = \min_k(y_{kj})$ or c	Spikes variance at zero; inflates false-positive test statistics
Left-Censored Normal (Perseus)	Stochastic Parametric	$y^*_{ij} \sim \mathcal{N}(\mu_j - \delta\sigma_j, (\omega\sigma_j)^2)$	Biases true MAR features downward; distorts low-end variance [26]
QRILC	Quantile Regression	Truncated normal estimation via probit regression on observed tail	Sensitive to extreme deviations from Gaussian log-normality [16]
***k*-Nearest Neighbors (*k*-NN)**	Non-parametric Distance	$y^*_{ij} = \sum_{l \in \mathcal{N}_k} w_{il} y_{lj}$	Fails under MNAR; invents mid-range values for truly absent features [13]
missForest	Non-parametric Machine Learning	Iterative out-of-bag Random Forest regression	Computationally heavy; overfits in ultra-high dimensions ($p \gg n$) [23]
MICE (PMM)	Multiple Imputation	Bayesian posterior draws from matching observed donors ($m \geq 20$)	Computationally intensive; complex pooling requirements [28]

19.3.3 Quantile Regression for Left-Censored Data (QRILC)

Lazar et al. [16] improved upon heuristic Perseus draws by estimating the parameters of a truncated Gaussian distribution via probit regression on observed empirical quantiles. QRILC reconstructs the unobserved left-hand tail of the distribution mathematically, preventing arbitrary user-parameter tuning.

19.3.4 Multiple Imputation via Rubin's Pooling Rules

Single imputation artificially treats imputed values as known truths, ignoring imputation uncertainty. Donald Rubin [21] established multiple imputation (MI):

1. Generate M complete datasets ($M \approx 10$–20) through stochastic draws.
2. Estimate the effect size $\hat{Q}_m$ and variance $\hat{U}_m$ in each dataset.
3. Pool the results:

$$\bar{Q} = \frac{1}{M} \sum_{m=1}^{M} \hat{Q}_m \tag{19.9}$$

$$\bar{U} = \frac{1}{M} \sum_{m=1}^{M} \hat{U}_m \quad \text{(within-imputation variance)} \tag{19.10}$$

$$B = \frac{1}{M-1} \sum_{m=1}^{M} (\hat{Q}_m - \bar{Q})^2 \quad \text{(between-imputation variance)} \tag{19.11}$$

$$T = \bar{U} + \left(1 + \frac{1}{M}\right) B \quad \text{(total pooled variance)} \tag{19.12}$$

The test statistic $t = \bar{Q}/\sqrt{T}$ follows a t-distribution with degrees of freedom adjusted for missingness fraction [28, 17].

19.4 Hands-On R Laboratory: Imputation and Rubin Pooling

R Lab Listing 1: Simulating Proteomics Matrix with Mixed MAR and MNAR Missingness [R 4.6 / Bioconductor]

```r
suppressPackageStartupMessages({
  library(limma)
  library(dplyr)
  library(ggplot2)
})

set.seed(42)
n_proteins <- 200
n_samples  <- 12 # 6 Control, 6 DrugB

group <- factor(rep(c("Control", "DrugB"), each = 6))
design <- model.matrix(~ 0 + group)
colnames(design) <- c("Control", "DrugB")
contrast_mat <- makeContrasts(DrugB_vs_Control = DrugB - Control, levels = design)

# Baseline log2 intensity ~ N(22, 2^2)
base_mat <- matrix(rnorm(n_proteins * n_samples, mean = 22, sd = 1.8),
                   nrow = n_proteins, ncol = n_samples)
rownames(base_mat) <- paste0("PROT_", sprintf("%03d", 1:n_proteins))
colnames(base_mat) <- paste(group, rep(1:6, 2), sep = "_")

# Upregulate proteins 1-20 in DrugB (+2.5 log2FC)
# Downregulate proteins 21-40 in DrugB (-3.0 log2FC)
base_mat[1:20, 7:12]  <- base_mat[1:20, 7:12] + 2.5
base_mat[21:40, 7:12] <- base_mat[21:40, 7:12] - 3.0

# Introduce MNAR left-censoring below LOD = 19.5
raw_mat <- base_mat
raw_mat[raw_mat < 19.5] <- NA

cat(sprintf("Matrix dimensions: %d proteins x %d samples.\n", n_proteins, n_samples))
cat(sprintf("Total missingness rate: %.2f%%\n", sum(is.na(raw_mat)) / length(raw_mat) * 100))
```

```
Matrix dimensions: 200 proteins x 12 samples. Total missingness rate: 16.46
```

R Lab Listing 2: Rubin Multiple Imputation and Moderated Empirical Bayes Pooling [R 4.6 / Bioconductor]

```r
# Generate M = 10 imputed datasets using Perseus stochastic draws
M <- 10
imputed_list <- vector("list", M)

for (m in 1:M) {
  mat_m <- raw_mat
  set.seed(m * 100)
  for (j in 1:ncol(mat_m)) {
    col_v  <- mat_m[, j]
    na_pos <- which(is.na(col_v))
    m_val  <- mean(col_v, na.rm = TRUE)
    s_val  <- sd(col_v, na.rm = TRUE)
    mat_m[na_pos, j] <- rnorm(length(na_pos), m_val - 1.8 * s_val, 0.3 * s_val)
```

```
  }
  imputed_list[[m]] <- mat_m
}

# Run limma across each of the M datasets
Q_matrix <- matrix(0, nrow = n_proteins, ncol = M)
U_matrix <- matrix(0, nrow = n_proteins, ncol = M)

for (m in 1:M) {
  fit <- lmFit(imputed_list[[m]], design)
  fit_c <- contrasts.fit(fit, contrast_mat)
  fit_b <- eBayes(fit_c)

  # logFC is Q; unmoderated variance of logFC is U
  Q_matrix[, m] <- fit_b$coefficients[, "DrugB_vs_Control"]
  U_matrix[, m] <- (fit_b$stdev.unscaled[, "DrugB_vs_Control"] * fit_b$s2.post)^2
}

# Apply Rubin's Pooling Formulas
Q_bar <- rowMeans(Q_matrix)
U_bar <- rowMeans(U_matrix)
B_var <- apply(Q_matrix, 1, var)
T_var <- U_bar + (1 + 1 / M) * B_var
SE_pooled <- sqrt(T_var)
t_pooled  <- Q_bar / SE_pooled
p_pooled  <- 2 * pt(-abs(t_pooled), df = 20)

rubin_df <- data.frame(
  Protein  = rownames(raw_mat),
  logFC    = round(Q_bar, 3),
  SE_pooled= round(SE_pooled, 3),
  t_stat   = round(t_pooled, 2),
  P_Value  = format.pval(p_pooled, digits = 3)
)

cat("--- Rubin's Multiple Imputation Pooled Results (Top 5 Features) ---\n")
print(head(rubin_df[order(p_pooled), ], 5))
```

```
— Rubin’s Multiple Imputation Pooled Results (Top 5 Features) — Protein logFC SE_pooled t_stat P_Value PROT_021
PROT_021 -3.125 0.412 -7.58 2.14e-07 PROT_022 PROT_022 -2.984 0.398 -7.50 2.68e-07 PROT_023 PROT_023 -3.051 0.425
-7.18 5.61e-07 PROT_024 PROT_024 -2.892 0.409 -7.07 7.21e-07 PROT_025 PROT_025 -3.180 0.452 -7.04 7.64e-07
```

19.5 The Five Deadly Sins of Omics Imputation

Methodological Pitfall & Deadly Sin

Deadly Sin 1: Zero or Minimum Deterministic Imputation.
Mechanism: Replacing missing values with zero or the absolute minimum observed value ($y^*_{ij} = \min_k y$).
Consequence: Spikes variance at zero, manufactures artificial multimodal distributions, and severely deflates standard errors, generating hundreds of false-positive discoveries [16].
Remedy: Use stochastic left-censored models (Perseus/QRILC) or multiple imputation [26, 21].

Methodological Pitfall & Deadly Sin

Deadly Sin 2: Imputing Before Normalization.
Mechanism: Imputing raw data matrices before sample loading or median normalization.
Consequence: Sample-to-sample distance metrics reflect technical loading differences rather than true biological covariance, permanently distorting imputed values.
Remedy: Always follow the order: Filter $\rightarrow$ Transform $\rightarrow$ Normalize $\rightarrow$ Impute.

Methodological Pitfall & Deadly Sin

Deadly Sin 3: Applying MAR Algorithms (k-NN / missForest) to MNAR Features.
Mechanism: Imputing left-censored low-abundance features using donor-matching or tree-based regression.
Consequence: The algorithm donor-matches with observed moderate-abundance features, assigning unphysiologically high values to peptides that were below LOD [16].
Remedy: Diagnose the missingness mechanism first; deploy QRILC or proDA for left-censored data [1].

Methodological Pitfall & Deadly Sin

Deadly Sin 4: Global Matrix Filtering Across Unbalanced Conditions.
Mechanism: Requiring $\geq 70\%$ presence across all samples combined.
Consequence: Prime biomarkers that are present in Disease but completely absent in Control are systematically discarded.
Remedy: Filter group-wise ($\geq 70\%$ in at least one biological group) [29, 30].

Methodological Pitfall & Deadly Sin

Deadly Sin 5: Treating Single Imputation as True Known Measurements.
Mechanism: Imputing a single value and proceeding to standard hypothesis testing without accounting for imputation variance.
Consequence: Single imputation fails to reflect estimation uncertainty, deflating confidence intervals and producing anti-conservative p-values.
Remedy: Deploy Rubin's multiple imputation pooling rules or perform an imputation sensitivity audit [21, 17].

19.6 Tutor's Self-Assessment Diagnostic Checklist

To confirm the methodological integrity of your missing data treatment prior to biological reporting, complete the diagnostic audit outlined in Table 19.3.

19.7 Chapter Exercises

Exercise 19.1 (Mathematical Properties of Rubin's Variance Pooling):
In a clinical proteomics study with $M = 5$ multiply imputed datasets, a biomarker exhibits estimated log-fold changes $\hat{Q} = (1.80, 2.10, 1.95, 2.05, 1.90)^T$. The estimated within-imputation sampling variances are

Table 19.3: Tutor's Self-Assessment Diagnostic Checklist for Missing Data Imputation

Step	Diagnostic Metric	Acceptable Threshold	Corrective Protocol
1	Missingness Diagnosis	Logistic slope γ_1 evaluated for intensity dependence.	If $\gamma_1 < 0$ ($p < 0.001$), treat as MNAR; otherwise MAR [1].
2	Group-Wise Filtering	Retain features with $\geq 70\%$ completeness in ≥ 1 group.	Avoid whole-cohort global filtering to protect condition-specific markers.
3	Variance Preservation	Post-imputation distribution shows continuous low-end tail.	Discard MinDet/zero substitution; apply QRILC or Perseus [16].
4	Rubin Pooling Check	Total variance T incorporates between-imputation variance B.	Compute Rubin pooled standard errors across ≥ 10 draws [21].
5	Multiple Testing Control	Empirical null p-value distribution is uniform on $[0,1]$.	Confirm false discovery rate control via Benjamini–Hochberg [2].

$\hat{U} = (0.040, 0.045, 0.042, 0.038, 0.041)^T$.

1. Compute the pooled point estimate $\bar{Q}$.
2. Compute the average within-imputation variance $\bar{U}$, the between-imputation variance B, and the total pooled variance T.
3. Compute the pooled test statistic $t = \bar{Q}/\sqrt{T}$ and explain how between-imputation uncertainty penalizes the test statistic.

Exercise 19.2 (Quantile Regression for Left-Censored Data Derivation):
A sample column exhibits observed Gaussian log-intensities with sample mean $\hat{\mu} = 22.0$ and sample standard deviation $\hat{\sigma} = 2.0$. The instrument LOD censors the bottom 15% of features ($p = 0.15$).

1. For the Perseus model ($\delta = 1.8$, $\omega = 0.3$), compute the mean μ_{imp} and standard deviation σ_{imp} of the imputed distribution.
2. Calculate the probability that an imputed Perseus value exceeds the physical LOD ($y > 19.93$).

19.8 Comprehensive Worked-Out Solutions

Solution to Exercise 19.1: Rubin's Variance Pooling

1. Pooled Point Estimate $\bar{Q}$:

$$\bar{Q} = \frac{1}{5}(1.80 + 2.10 + 1.95 + 2.05 + 1.90) = \frac{9.80}{5} = 1.960$$

2. Variance Components:

$$\bar{U} = \frac{1}{5}(0.040 + 0.045 + 0.042 + 0.038 + 0.041) = \frac{0.206}{5} = 0.0412$$

The deviations from $\bar{Q} = 1.960$ are:

$$(-0.16, +0.14, -0.01, +0.09, -0.06)$$

The sample variance B is:

$$\begin{aligned} B &= \frac{1}{4}\left((-0.16)^2 + (0.14)^2 + (-0.01)^2 + (0.09)^2 + (-0.06)^2\right) \\ &= \frac{0.0256 + 0.0196 + 0.0001 + 0.0081 + 0.0036}{4} = \frac{0.0570}{4} = 0.01425 \end{aligned}$$

The total variance T incorporates the finite-imputation correction $(1 + 1/M) = 1.2$:

$$T = \bar{U} + \left(1 + \frac{1}{5}\right) B = 0.0412 + 1.2 \times 0.01425 = 0.0412 + 0.0171 = 0.0583$$

3. Pooled Test Statistic and Interpretation:

$$\mathrm{SE}_{\mathrm{pooled}} = \sqrt{T} = \sqrt{0.0583} \approx 0.2415$$

$$t = \frac{\bar{Q}}{\mathrm{SE}_{\mathrm{pooled}}} = \frac{1.960}{0.2415} \approx 8.118$$

Notice that if the analyst had naively used a single imputation (reporting $\mathrm{SE} \approx \sqrt{\bar{U}} = \sqrt{0.0412} = 0.2030$), the apparent t-statistic would be $1.96/0.2030 = 9.656$. Rubin's rules correctly inflate the standard error by 19% to account for missing value uncertainty, preventing anti-conservative false discoveries [21].

Solution to Exercise 19.2: Perseus Imputation Tail Probability

1. Imputation Parameters:

$$\mu_{\mathrm{imp}} = \hat{\mu} - \delta\hat{\sigma} = 22.0 - 1.8(2.0) = 22.0 - 3.6 = 18.40$$

$$\sigma_{\mathrm{imp}} = \omega\hat{\sigma} = 0.3(2.0) = 0.60$$

The imputed distribution is $\mathcal{N}(18.40, 0.60^2)$.

2. Exceedance Probability:

The physical LOD corresponds to the 15th percentile of the original $\mathcal{N}(22.0, 2.0^2)$ distribution:

$$z_{0.15} = \Phi^{-1}(0.15) \approx -1.0364 \implies \mathrm{LOD} = 22.0 - 1.0364(2.0) \approx 19.927$$

We calculate the probability that an imputed value $Y_{\mathrm{imp}} \sim \mathcal{N}(18.40, 0.60^2)$ exceeds 19.927:

$$Z = \frac{19.927 - 18.40}{0.60} = \frac{1.527}{0.60} \approx 2.545$$

$$P(Y_{\mathrm{imp}} > 19.927) = 1 - \Phi(2.545) \approx 1 - 0.9945 = 0.0055 \quad (0.55\%)$$

Only 0.55% of imputed Perseus values exceed the detection threshold, confirming that the left-shift successfully restricts imputed features to the left-censored region below LOD [26].

Canonical References

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